

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
df = pd.read_csv("Iris.csv", header=0)
print(df.head(n=5))
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
df.isnull()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
145	False	False	False	False	False	False
146	False	False	False	False	False	False
147	False	False	False	False	False	False
148	False	False	False	False	False	False
149	False	False	False	False	False	False

150 rows × 6 columns

```
df.shape
```

```
(150, 6)
```

```
df.size
```

```
900
```

```
df.columns
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
      'Species'],  
      dtype='object')
```

```
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: [Generate code with df](#)

[New interactive sheet](#)

```
df.tail()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	
145	146	6.7	3.0	5.2	2.3	Iris-virginica	
146	147	6.3	2.5	5.0	1.9	Iris-virginica	
147	148	6.5	3.0	5.2	2.0	Iris-virginica	
148	149	6.2	3.4	5.4	2.3	Iris-virginica	
149	150	5.9	3.0	5.1	1.8	Iris-virginica	

```
df.dtypes
```

	0
Id	int64
SepalLengthCm	float64
SepalWidthCm	float64
PetalLengthCm	float64
PetalWidthCm	float64
Species	object

```
dtype: object
```

```
df.isnull().sum()
```

	0
Id	0
SepalLengthCm	0
SepalWidthCm	0
PetalLengthCm	0
PetalWidthCm	0
Species	0

```
dtype: int64
```

```
df.isna()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	
...	...	...	...	...	...	...	
145	False	False	False	False	False	False	
146	False	False	False	False	False	False	
147	False	False	False	False	False	False	
148	False	False	False	False	False	False	
149	False	False	False	False	False	False	

```
150 rows × 6 columns
```

```
df.sample()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	
100	101	6.3	3.3	6.0	2.5	Iris-virginica	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0    Id              150 non-null    int64
1    SepalLengthCm   150 non-null    float64
2    SepalWidthCm    150 non-null    float64
3    PetalLengthCm   150 non-null    float64
4    PetalWidthCm    150 non-null    float64
5    Species         150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
df['Id']
```

	Id
0	1
1	2
2	3
3	4
4	5
...	...
145	146
146	147
147	148
148	149
149	150

150 rows × 1 columns

dtype: int64

```
df[['Id', 'SepalLengthCm']]
```

	Id	SepalLengthCm	
0	1	5.1	
1	2	4.9	
2	3	4.7	
3	4	4.6	
4	5	5.0	
...	...	...	
145	146	6.7	
146	147	6.3	
147	148	6.5	
148	149	6.2	
149	150	5.9	

150 rows × 2 columns

```
df.iloc[0]
```

	0
Id	1
SepalLengthCm	5.1
SepalWidthCm	3.5
PetalLengthCm	1.4
PetalWidthCm	0.2
Species	Iris-setosa

dtype: object

```
df.iloc[3:4]
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
3	4	4.6	3.1	1.5	0.2	Iris-setosa

```
from sklearn import preprocessing
```

```
x = df['SepalLengthCm'].values.astype(float)
```

```
min_max_scaler = preprocessing.MinMaxScaler()
```

```
x = x.reshape(-1, 1)
x_scaled = min_max_scaler.fit_transform(x)
```

```
df_normalized = pd.DataFrame(x_scaled)
```

```
df_normalized
```

	0
0	0.222222
1	0.166667
2	0.111111
3	0.083333
4	0.194444
...	...
145	0.666667
146	0.555556
147	0.611111
148	0.527778
149	0.444444

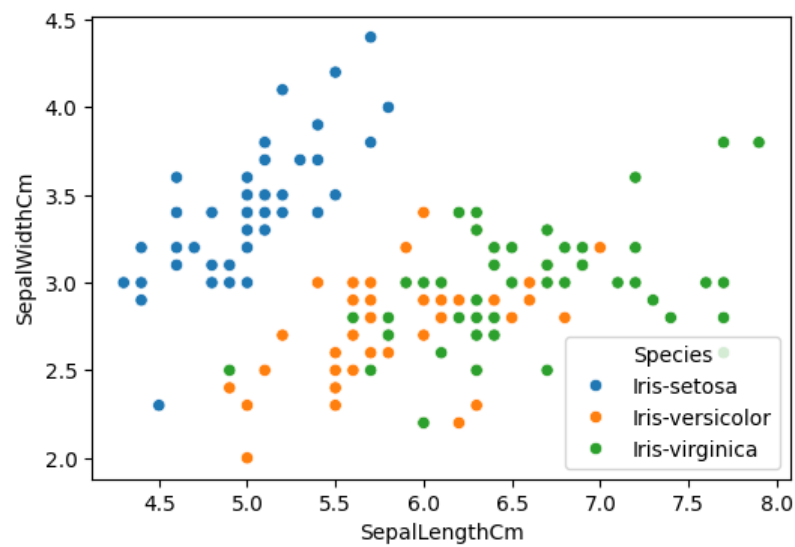
150 rows × 1 columns

Next steps:

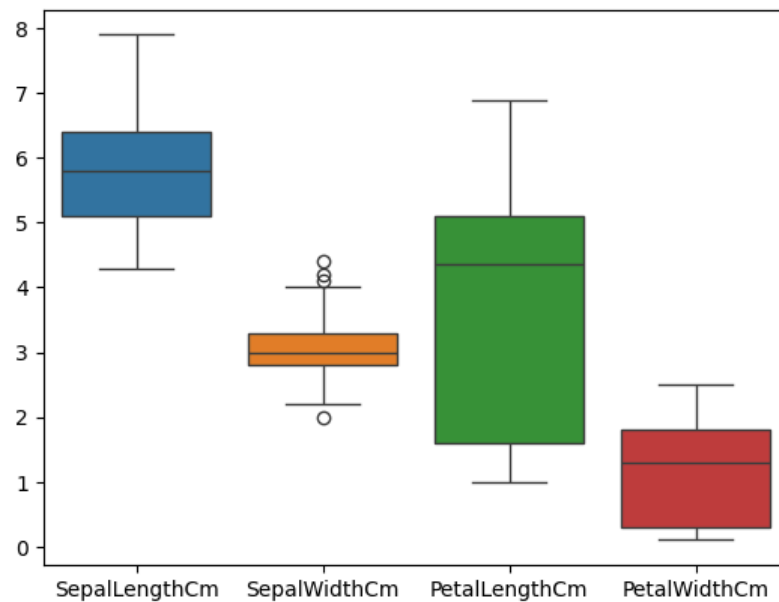
[Generate code with df_normalized](#)
[New interactive sheet](#)

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='Species', data=df)
plt.show()
```



```
sns.boxplot(data=df.drop('Id',axis=1))  
plt.show()
```



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